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Mitigation of severe accidents in AREVA's Gen 3+ nuclear power plants

M. Fischer*, A. Henning, R. Surmann

AREVA, E&P Division, Erlangen 91058, Germany

ABSTRACT

The current AREVA Gen 3+ PWR designs (EPR™ and ATMEA1) are based on the proven defense-in-depth safety concepts inherited from their predecessors, the French "N4" and the German "Konvoi" reactors. Complemented by specific enhancements, including higher redundancy and diversity as well as the use of passive systems, this leads to very low values of the core damage frequency (CDF). Notwithstanding this very low probability, dedicated design measures have been implemented to improve the response of the plant in case of a postulated severe accident (SA) with core melting. This way not only the frequency of large-early-releases (LERF) but also the related radiological consequences are drastically reduced. Situations that potentially lead to high loads that can challenge the short-term integrity of the containment, like RPV melt-through under high pressure, energetic hydrogen/steam explosions, as well as long-term containment failure caused by internal over-pressure are avoided by a combination of preventive measures and dedicated systems.

At the example of the EPR™, the paper gives an overview of the severe accident mitigation strategy and the related measures and systems of AREVAs current Gen 3+ reactors, with special focus on the function of the core melt stabilization system.

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1. Introduction

All of AREVAs advanced Gen 3+ reactor designs, the EPR™ and ATMEA1 PWRs as well as the KERENA BWR, give extended consideration to the prevention of severe accidents. Despite the achieved very low values of the core melt frequency, additional measures capable of drastically reducing the environmental impact in case a severe accident (SA) are implemented in the design. Their target is to eliminate the need for emergency evacuations of the surrounding population and for long-term restrictions with respect to the consumption of locally grown food, even in case of such a highly unlikely event. The adopted safety concepts meet advanced regulatory requirements, including IAEA safety guide, Technical Guidelines (GPR/German experts), and SECY 93-087 (NRC).

The SA mitigation concept is illustrated at the example of the EPR™ PWR, a 4-loop reactor with a power of about 4650 MWth, currently under construction in Finland, France and the P.R. China.

2. EPR™ severe accident mitigation strategy overview

A severe accident commences when loss of emergency core cooling functions have failed to maintain the core in a coolable and stable configuration. For this case, in order to achieve the international consequence targets, the EPR™ is equipped with dedicated severe accident mitigation measures that are 'novel' to existing PWRs. These measures, see Fig. 1, aim at preserving the containment integrity not only during a limited time but throughout the SA. To achieve this target a balanced approach adequately considering all related challenges and containment failure modes is followed.

Short-term situations potentially leading to high thermal and mechanical loads on the containment, like RPV high pressure melt-through or energetic hydrogen and steam explosion are excluded by adequate preventive measures. For the resulting low pressure sequences, mid- and long-term containment failure caused by over-pressurization or basemat melt-through is avoided by a combination of design measures and dedicated systems.

For the proof of function of these mostly passive systems a wide spectrum of initiating events and scenarios, including Large Break (LB)-LOCA, Small Break (SB)-LOCA, Loss of Offsite Power (LOOP), and Total Loss of Feed Water (TLOFW), combined with a successful depressurization of the primary circuit, are considered. The related demonstration is provided on a deterministic basis, applying best-estimate assumptions and methods. During the SA, dedicated instrumentation allows monitoring the course of events and deviation from the predefined mitigation path.

The robustness of the SA mitigation system and its ability to preserve containment leak-tightness is also demonstrated under bounding conditions which include elevated RPV failure pressure or internal late reflood. Scenarios with a postulated complete failure to depressurize the primary system are not part of the SA mitigation system design base and remain residual risk to be evaluated as part of PSA level-2.

3. Prevention of high pressure RPV failure

Melt-through of the RPV under high system pressure involves the possibility of energetic missile formation as well as High Pressure Melt Ejections (HPME) leading to Direct Containment Heating (DCH). Because of the related risk of short-term containment

* Corresponding author.
E-mail address: manfred.fischer@areva.com (M. Fischer).

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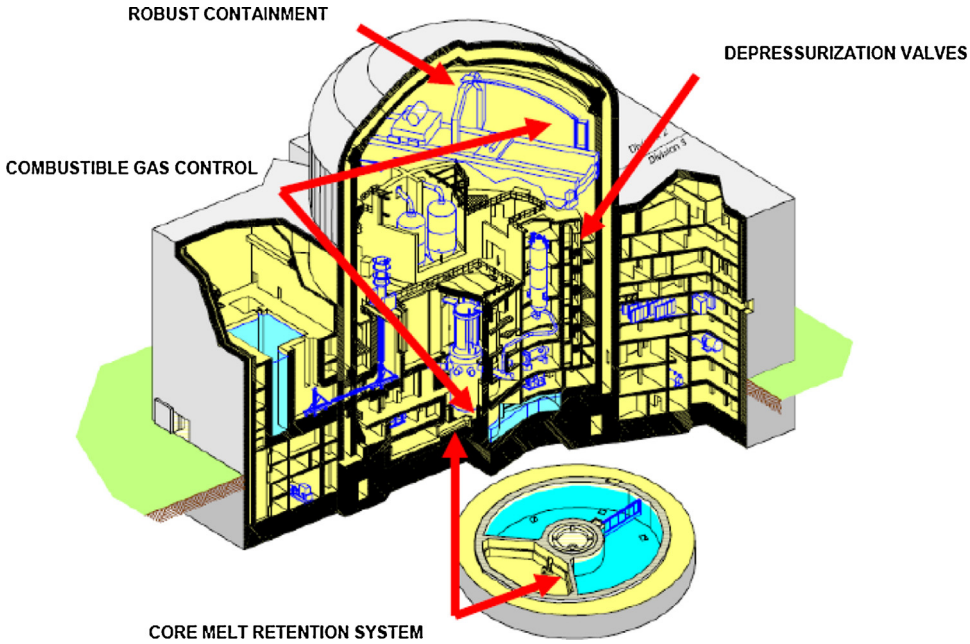

Fig. 1. EPR™ plant scheme, with severe accident mitigation measures.

failure, the reliable transfer of high pressure into low pressure core melt sequences is of paramount importance for preserving the containment integrity in case of SA. To ensure successful depressurization, the EPR™ provides two redundant lines, each with two dedicated high-capacity Pressurizer Dedicated Safety (PDS) valves in series, in addition to the existing three pressurizer safety valves (PSV), see Fig. 2. The PDS valves are manually activated, at the latest when the core outlet temperature exceeds 650 (C. The sizing of the PDS valves ensures rapid reduction of the primary system pressure below 5 bar. Any later pressure increase in the primary circuit is only possible as a consequence of active late reflood.

High redundancy for opening the valves is achieved by associating their electrical and I&C systems to all 4 plant divisions. The required power to open can be alternatively provided by On- and Off-site Electrical Systems, including SBO diesels, as well as by two long-duration (>12 h) dedicated SA batteries. The PDS and PSV valves feed into a 40 m^3 pressure relief tank (PRT), which – after failure of a 20 bar rupture disk – discharges into two out of the four reactor coolant pump (RCP) rooms. Direct release into the containment was preferred against condensation in the IRWST because of the positive effect on combustible gas mixing and inertization.

4. Prevention of energetic steam explosions

A steam explosion can evolve from situations with an intense interaction and mixing between molten fuel and water. Pressure waves passing through the water can then disrupt the vapor film and cause direct contact between the hot surface of the particles and the water and trigger a rapid synchronous exchange of energy that amplifies the primary wave. Given a sufficiently long interaction period, this process can continue until a highly energetic shock wave is formed. Steam explosions are especially strong if the water pool is subcooled and in the absence of non-condensable gases.

A steam explosion may happen at different stages in the progression of a severe accident, most notably, when melt drains from the disintegrating core into the residual water in the reactor lower

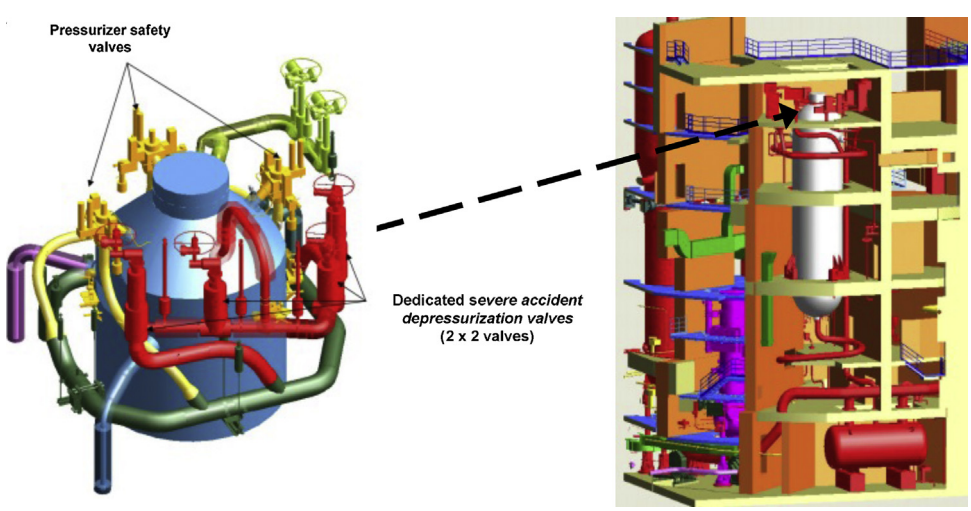

Fig. 2. EPR™, positions of PSV/PDS valves and PRT tank, relative to the pressurizer tank.

head but also after vessel melt-through, should the region below be flooded with water.

4.1. In-vessel steam explosion

No specific countermeasures against in-vessel steam explosions are taken in the EPR™ design. This is because the probability of an in-vessel steam explosion being sufficiently energetic to damage the RPV upper or lower head can be deterministically shown to be very low. This demonstration is based on best-estimate views on melt progression/relocation and conversion efficiencies and supported by modeling and experiments. The obtained results are consistent with the view of the OECD SERENA-1 project, which concludes that this mode of vessel failure can be considered resolved from a risk perspective, meaning that it is of very low likelihood and of little or no significance to the overall risk from a nuclear power plant.

4.2. Ex-vessel steam explosion

The EPR™ SA mitigation strategy does not intent to retain the molten core inside the vessel. Instead, an ex-vessel core catcher is provided in an adjacent, isolated compartment. As both, the reactor pit and the core catcher, are initially dry there is no risk of energetic ex-vessel steam explosions resulting from melt entering into a water pool. Unintended leakage or water ingress into the core catcher during operation can be detected and terminated. Flooding of the melt after dry spreading into the core catcher is performed passively at low flow rate and during melt interaction with sacrificial concrete. Promoted by the related superficial agitation of the melt pool, all added water will initially evaporate so that a deep water pool atop the melt can only form after the melt's surface is no longer liquid but covered by a solid crust. There is a large experimental database demonstrating the absence of steam explosions under the conditions of top flooding during molten core concrete interaction (MCCI).

5. Prevention of basemat melt-through

5.1. In-vessel- vs. ex-vessel melt retention

In-vessel retention (IVR) attempts to stabilize the molten core within the RPV by sustained cooling of the outside of the vessel. This solution is attractive as it minimizes the dispersal of melt and requires only limited additional hardware. The success of IVR is related to the question: does the heat flux to the water – at any time – stay below the local "Critical Heat Flux" (CHF) at the outside of the vessel? This is important because a transition to film boiling can lead to a fast and strong increase in the local RPV wall temperature potentially followed by creep failure or melt-through, if the affected area is large enough.

The validation of IVR thus requires a thermal analysis of the combined system of melt, RPV wall, internal structures, and external 2-phase flow. The heat flux to the inner surface of the RPV depends on the natural convection heat flux distribution inside the molten pool and on the formation and spatial arrangement of stratified metallic and oxidic melt layers. The fact that sideward heat transport by convection/conduction out of a molten metallic (steel) layer is more efficient than radiant heat transport off its free surface, gives rise to what is known as the "focusing effect". A low-density metallic phase atop the decay-heated oxide pool can therefore concentrate heat to its circumferential contact region with the vessel. For sufficiently thin layers, the "focused" heat flux can become very high and, by far, exceed the maximum heat flux inside the oxidic pool below.

This can be critical particularly for small-sized vessels and reactors with high thermal (and decay) power. The evolution of the thickness of the upper metallic layer is heavily affected by the melting sequence of the steel internals as well as by thermo-chemical processes that can create a dense, U-rich metallic phase suspended near the bottom of the oxidic pool (Fischer et al., 2011). As both processes depend on the peculiarities of the underlying accident, incl. the degree of Zr-oxidation, the thickness of the upper metal layer and the thermal loads on the vessel can only be predicted with high uncertainties. Considering that (i) low margins for IVR involve an additional risk of steam explosion-induced early containment failure after vessel melt-through, in combination with the fact that (ii) in the likely case of an open primary circuit the long-lasting very high temperatures in the (dry?) in-vessel melt pool can cause significant activity release into the containment, IVR is not applied for both the EPR™ and ATMEA1. The chosen strategy is therefore a controlled stabilization of the molten core in an ex-vessel core catcher. However, IVR is applied for AREVAs KERENA BWR (Fischer and Levy, 2010), because here the large size of the vessel, the long grace period before core melting, and the large internal steel masses ensure safe margins.

5.2. Core melt stabilization strategy overview

The core melt stabilization system (CMSS) of the EPR™, see Fig. 1, prevents the interaction between the molten core and the basemat and thus eliminates the risk of containment failure caused by:

- the penetration of the embedded containment liner
- the long-term heat-up and mechanical deformation of the containment civil structure
- the sustained release of non-condensable gas into the containment

The key component of the CMSS is the core catcher (CC), a large metallic crucible into which the melt spreads and where it is quenched and cooled. The CC is located in a dedicated compartment lateral to the reactor pit, see Fig. 3. Due to its spatial and functional separation from the pit, the core catcher is decoupled from the loads related with the initial RPV failure, which are now absorbed by the robust structures located in the reactor pit. In addition, even an unintended flooding of the core catcher during power operation does not lead to an imminent safety risk.

Gravity-driven melt relocation into the core catcher is preceded by a phase of temporary melt retention in the reactor pit. During this phase the already released melt interacts with the sacrificial concrete (at bottom and sides) and with the residual lower head (above). This ensures incorporation of all core material into the molten pool, significantly before the concrete melt plug in the center of the pit bottom is reached and penetrated.

Once this ultimately happens, the melt flows via an embedded melt discharge channel into the lateral core catcher where it activates the opening of passive flooding valves that start the overflow of water from the Internal Refueling Water Storage Tank (IRWST). After filling the space around the CC, the water eventually pours onto the surface of the spread melt from the top. At the end of this quench period, the melt is completely surrounded by water, or cooled structures and all decay heat produced inside the melt is extracted by heat-up and evaporation of the water.

Along the described sequence, the transformation of the molten core into a coolable and permanently cooled configuration is achieved on the basis of simple physics and without requiring either operator action, or the use of internal or external active systems.

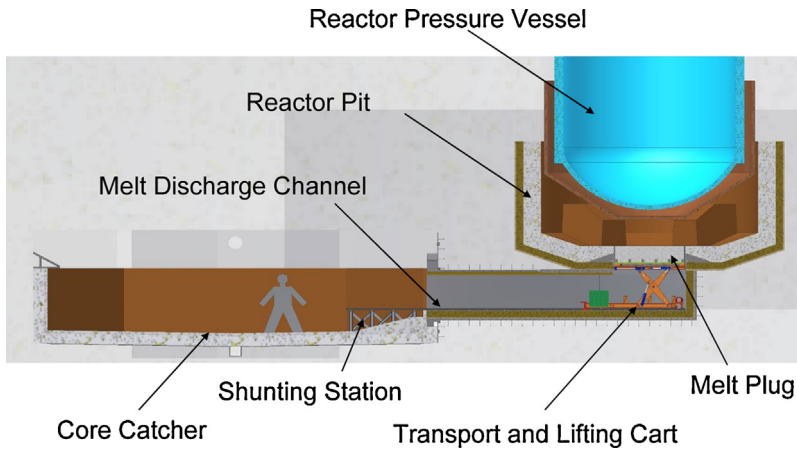

Fig. 3. EPR™ core melt stabilization system, main components.

5.3. Controlled state approach

No general prediction on when and how the core will degrade, melt and finally penetrate the vessel in a severe accident is possible, since the course of these in-vessel events depends on several unknowns, like:

- the state and history of the core (bundle characteristics, burn-up, etc.);
- the details of the preceding accident scenario (initiating event, operator response, etc.);
- stochastic processes during core degradation/melting (like material relocation, crust break-up, etc.).

In addition, there exist uncertainties when describing the underlying thermal, mechanical and chemical processes, in particular during the late in-vessel phase, before vessel failure.

These inherent uncertainties are addressed in the design of the EPR™ reactor pit as the related measures influence the course of events and strongly increase the predictability of the final situation. This makes all following stages of the mitigation path independent of the conditions at vessel failure and eliminates the necessity to analyze and predict them in detail.

By this "controlled state" approach, described in the following, intermediate states and conditions are generated that are sufficiently well defined and for which assured knowledge exists. This eases the validation, as only a limited spectrum of states and conditions needs to be considered for the proof of the core catcher function.

5.4. Melt accumulation and conditioning for spreading

In case of SA, the release of the molten material from the RPV will, most likely, not take place in one pour, but during a certain extended period of time. To ensure that the entire core inventory relocates into the core catcher in one event, the CMSS takes credit of a phase of temporary melt retention and accumulation in the pit. It is based on the provision of a layer of sacrificial material that must be penetrated by the melt prior to its escape from the pit.

The sacrificial concrete layer is completely surrounded by a melt-resistant ceramic layer, which protects the civil structure and compensates non-homogeneities in the concrete erosion process. The protective layer has only one penetration, namely at the position of the removable melt plug in the center of the pit bottom, which therefore acts as the predefined release location for the melt. The melt plug is shielded against mechanical impact of the detaching lower head by surrounding concrete bumpers.

After the initial RPV melt-through, the released material starts interacting with the sacrificial concrete. During the MCCI the heat flux emitted from the surface of the pool heats the RPV lower head from below. In addition, due to the narrow pit, there will be direct melt contact after about half of the core material is released. Heating from below is supplemented by the decay heat generated inside the vessel. Considering that:

- most of the upward transported heat will actually enter the RPV lower head
- downward heat transport in the MCCI pool is, at the best, as effective as upward or sideward heat transport
- the total downward heat transport is defined by the amount of provided sacrificial concrete

it can be shown that (i) the progression of the MCCI in downward direction is linked with the heat-up of the RPV lower head, and (ii) the amount of heat carried into the lower head is mainly determined by the thickness of the eroded sacrificial concrete and therefore widely independent of the decay power level (scenario) and the sequence of melt release from the RPV.

Whenever the lower head has absorbed a sufficient integral amount of heat, it is considered to collapse into the MCCI pool, carrying with it the remaining in-vessel fraction of the core. Corresponding analysis shows that, even under conservative assumptions, this will happen before half of the thickness of the provided sacrificial concrete layer is ablated.

Later on, most of the remaining sacrificial concrete will also become incorporated into the MCCI pool. The resulting, high concrete fraction dominates the state and properties of the melt at the end of the accumulation period and thus makes them well predictable.

The total duration of the retention period depends on the decay power level but is typically between 1 and 3 h. The end time is defined by the thermal destruction of the aluminum plate, the "gate", which forms the bottom of the melt plug. During its out-flow the melt steadily enlarges the initial hole, which again progressively accelerates the release. A sufficiently fluent state of the melt is ensured by the fact that, during the preceding MCCI, the pool must remain convective to carry the decay power to the surrounding colder interfaces.

The released melt is guided through a discharge channel into the core catcher, see Fig. 3. The channel surface is covered with a protective layer that withstands all related mechanical and thermal loads. Due to the fact that the channels exit level is above the

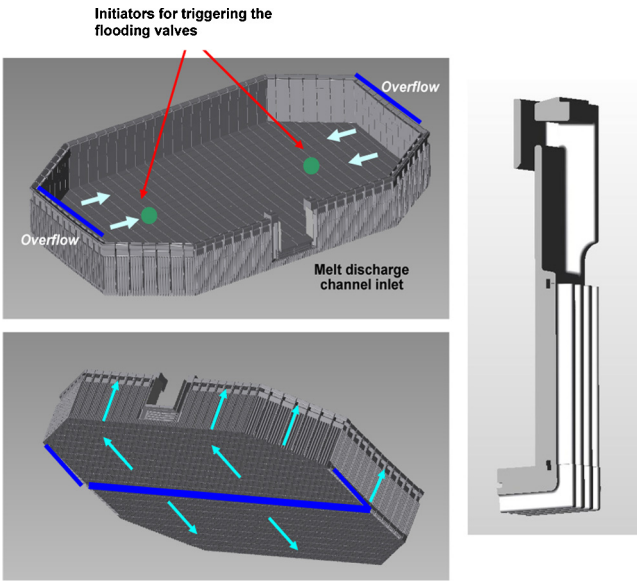

Fig. 4. EPR^{TM} core catcher structure (left) and individual lateral cooling element (right).

maximum melt level in the core catcher, practically all melt will be discharged.

5.5. Melt stabilization in the core catcher

The EPR^{TM} core catcher, see Fig. 4, is a massive cast iron structure weighing about 500 tons. It is assembled from a large number of individual elements connected with each other by a tongue-and-groove technique. The achieved structural flexibility avoids problems resulting from thermal deformation and non-homogeneous heat-up. The joints between the elements are closed by steel elements and compressible, temperature-resistant mineral wool inserts. The inner surface of the structure On the inside melt contact is delayed by is covered with concrete to protect the cooling elements from transient thermal loads during melt spreading.

The arrival of the melt triggers the opening of redundant, passive valves located in compartments lateral to the core catcher, thus initiating the gravity-driven overflow of water from the IRWST. Both, the arrival of the melt in the core catcher and the opening of the flooding valves, can be detected by SA-qualified instrumentation.

The incoming water first fills a central supply duct located underneath the core-catcher from where it rises into a system of parallel cooling channels formed by the fins of the bottom and sidewall cooling elements, see Fig. 4. Next, the water submerges the space behind the sidewall structure and finally pours onto the surface of the melt from the circumference after about 5 min. The outside of the cooling structure is completely submerged before being heated-up by the melt. On the inside, melt contact is delayed by the sacrificial concrete and by the high thermal inertia of the cooling elements. The generated steam will enter the containment via a vertical chimney that leads into compartments above the heavy floor.

As long as the IRWST contains enough water, the spread melt is kept submerged and cooled via passive overflow. In the long term the steaming rate will follow the course of decay power.

The efficiency of the core catcher is demonstrated based on analyses and experiments performed in an electrically heated 1:1 mockup of the cooling channel. The high thickness of the cooling structure results in a robust thermal behavior characterized by a low sensitivity against transient peaks after initial melt contact.

The thermal response of the cooling structure is analysed using measured data on heat transfer from prototypic spreading experiments. The conservative assumptions applied include zero fragmentation at the surface of the melt and the mixing of metal and oxide during MCCI. The design and technical realization of the EPR^{TM} CMSS are described in (Fischer and Henning, 2009).

6. Prevention of H2 combustions and explosions

Combustible gases, namely hydrogen, impose a threat for the containment integrity because of the dynamic pressure loads potentially resulting from energetic events (detonations) as well as from the quasi-static pressure and temperature build-up following slow burns, including standing flames. Detonations are of concern mainly during the early stage of the severe accident (in-vessel hydrogen production phase), when the hydrogen concentrations are high and the steam partial pressure is still low.

To control the hydrogen content in the containment atmosphere in case of SA, the EPR^{TM} is equipped with a dedicated Combustible Gas Control System (CGCS). It consists of two parts: (i) a passively activated and operating atmospheric mixing system which – early into the accident – transforms the former 2-room into a 1-room containment, and (ii) a large number of autocatalytic recombiners strategically placed in the containment and able to deplete hydrogen, even under steam-inert severe accident conditions. The combined functional requirements are to:

- limit the concentration in the containment to <10 vol.%, in order to avoid a global detonation
- allow higher concentrations than 10 vol.% only temporarily and only in small regions

During a SA, the success of these measures can be evaluated using data delivered by the Containment Atmosphere Monitoring Systems (CAMS).

The transformation from the 2-room (DBC1&2) into a 1-room containment (DBC3&4 and DEC) takes place passively by (i) the fast opening of large free flow cross-sectional areas at the pressure equalization ceiling, and (ii) the opening of additional inlets between annular rooms and equipment rooms near the IRWST level. As a consequence, convection flow is established throughout all relevant containment areas (equipment rooms, dome area, annular rooms, IRWST). Gas mixing is additionally supported by the low level of the PRT discharge lines, see Fig. 2, and the hot exhaust gas released from the autocatalytic recombiners, see Fig. 5.

The number and total capacity of the recombiners are chosen in consistence with the requests to:

- decrease the global H2-concentration in the containment below 4 vol% within a few hours, assuming a released H2 mass corresponding to 100% cladding oxidation.
- prevent a short-term global hydrogen concentration in the containment above 10 vol%.

In the long-term, practically all hydrogen is removed from the containment atmosphere by the installed recombiners, which eliminates the possibility of late hydrogen combustions. An overview of the function of the EPR^{TM} CGCS and the applied validation strategy is given in (Dimmelmeyer et al., 2010).

7. Prevention of containment overpressure failure

For containment p&T control, the EPR^{TM} is provided with a dedicated Containment Heat Removal System (CHRS) specifically qualified for SA conditions. For keeping the containment pressure

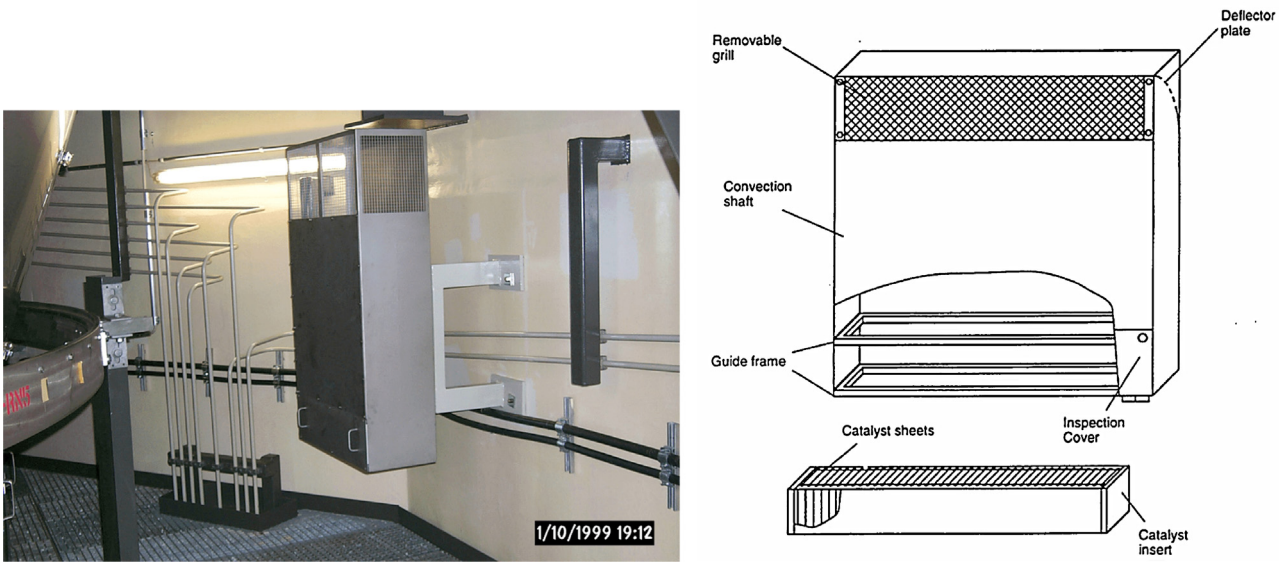

Fig. 5. AREVA autocatalytic H2-recombiners, when installed (left) and inner structure (right).

below the design pressure no activation of the CHRS is required during the minimum 24 h grace period, due to the large free volume of the EPR™ containment and the high heat storage capacity of its internal structures.

The CHRS takes suction from the IRWST through dedicated screens and re-injects the cooled-down water into the containment either via the spray system (which cools the atmosphere and condenses steam) or via the core melt spreading compartment.

All compartments housing components of the CHRS outside the containment are specifically protected and isolated and can thus be considered as an extension of the containment. The CHRS consists of two redundant trains and uses a separate cooling chain.

During the first days, the main target will be to reduce the containment pressure and to wash-out air-born fission products. During this period, spraying is the preferred CHRS mode. Later, one or both trains can be used to feed water directly into the spreading compartment. This lifts the water level above the overflow level to the heavy floor. From there, the water flows back into the IRWST, thus closing the loop for the CHRS (Fig. 6).

All decay heat then becomes removed by single-phase water flow instead of by evaporation and steam release into the containment. Therefore, ambient pressure can be reached without the need for venting. Nevertheless, a high-efficiency filtered venting system is available on request.

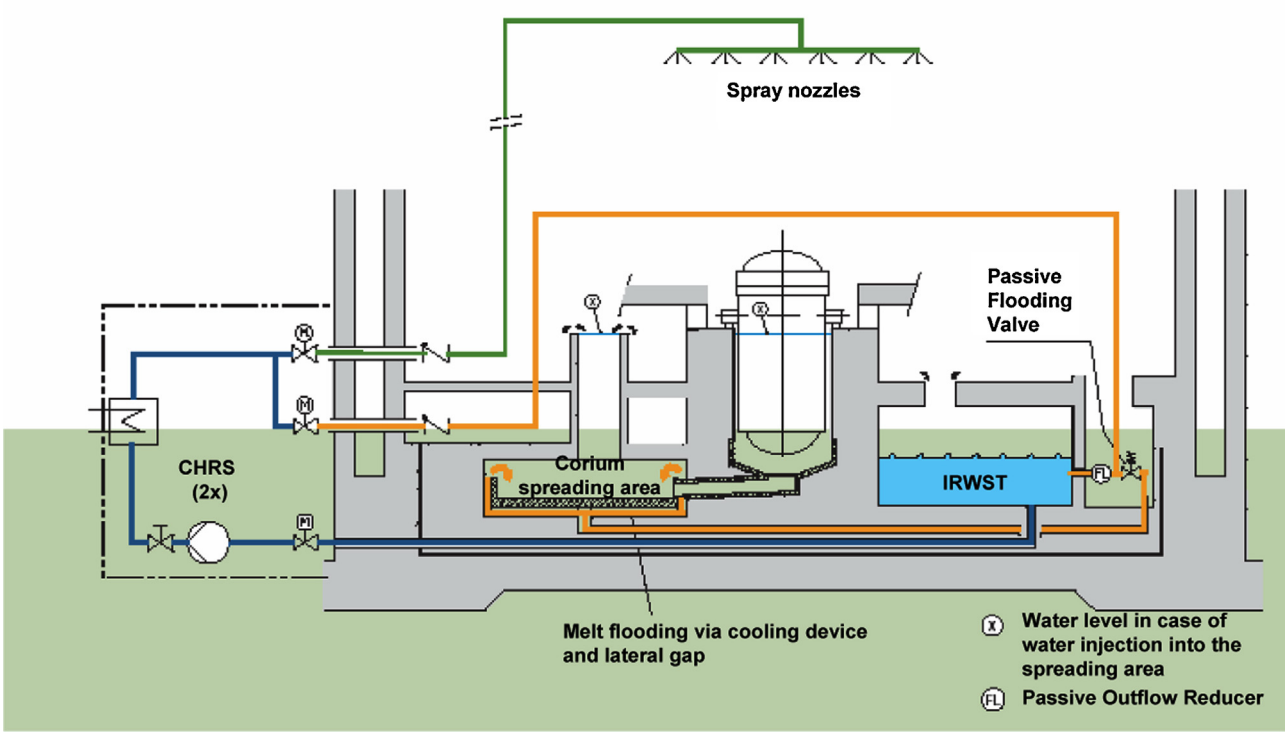

Fig. 6. CHRS flow scheme for spraying and direct injection into the core catcher.

In the described active injection mode, water will also enter the reactor pit and submerge the RPV up to the level of the loop-lines. This establishes long-term cooling for debris potentially remaining in the RPV, the pit, and the melt transfer channel.

8. Conclusion

The EPR™ design involves a complete and balanced set of components and systems for severe accident mitigation. They operate on physical principles that are simple and well understood. The potential impact of the remaining uncertainties is eliminated by a robust design of the components. The applied materials are commonly known and also used in other industrial applications.

The function of the SA systems is demonstrated for a wide spectrum of scenarios, based on analytical and experimental data and considering the state-of-the-art with respect to severe accident codes and the evolution of the core melt accident, incl. the behavior and the properties of the molten corium as well as its interaction with structural material.

The adequacy of the approach is reflected in the positive evaluation of the EPR™ SA concept in the EPR™ licensing procedures in Finland, France, China, the U.K., and the U.S.A. The ATMEA1 design takes benefit of these achievements by adopting these successfully validated solutions.

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